



ShowTXPD Tool

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Introduction

showTXPD is a useful for the user to visualize how the transmit beam is aligned with the pixel data. If the user wants to transmit a focused or a steered beam, with a certain delay, showTXPD can be used to confirm whether the TX and corresponding PData are defined correctly. This documentation will introduce different features of the showTXPD tool.

The user can call showTXPD with 0, 1, or 2 arguments. Call with no arguments to process all TX structures with first PData struct. If a range of TX values is desired, set the first argument to the range, e.g. showTXPD(1:10). If a PData struct other than one is to be specified, set the range of TX values and then specify the PData number in the 2nd argument, e.g. Run SetUpC5_2vWideBeamDoppler script, and then type showTXPD(61:63,2) on the command window. Only the TXPD of Doppler beam and the Region in PData(2) will be displayed. Multiple PData numbers are not allowed in the 2nd argument.

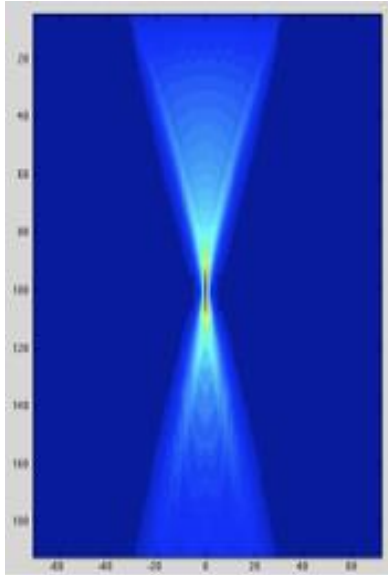
Features

Visualization of transmit beam profile

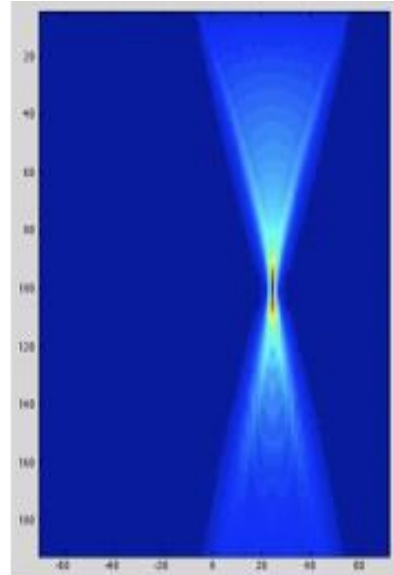
As shown in the Vantage Sequence Programming Manual, the showTXPD allows the user to visualize the transmit beam profile. Below are some examples:

Focused Beam:

TX(i).Origin = [0,0,0];

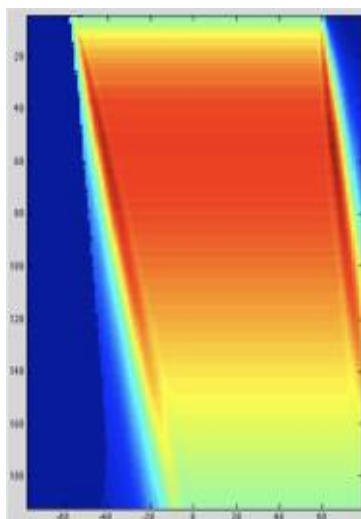


TX(i).Origin = [10,0,0];

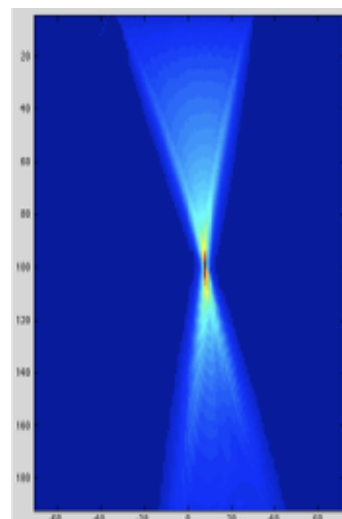


Steered Beam

Plane wave

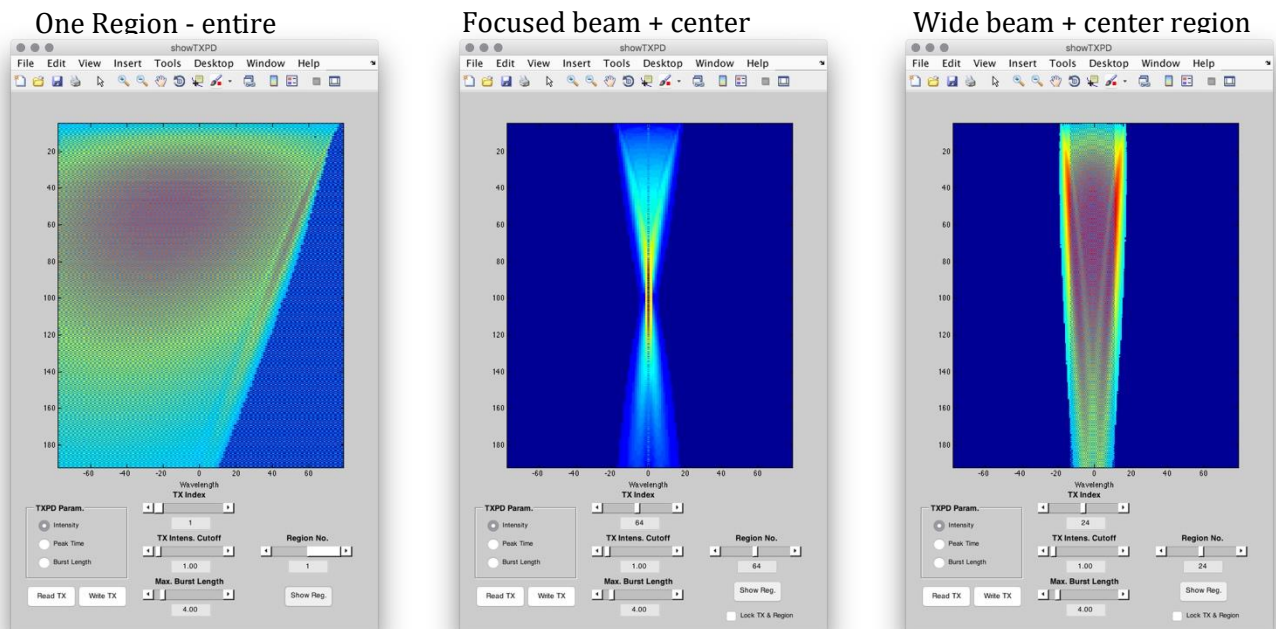


Focused beam



Visualization of PData.Region

The pixel data defined in the PData.Region will be displayed after invoking “Show Reg.”

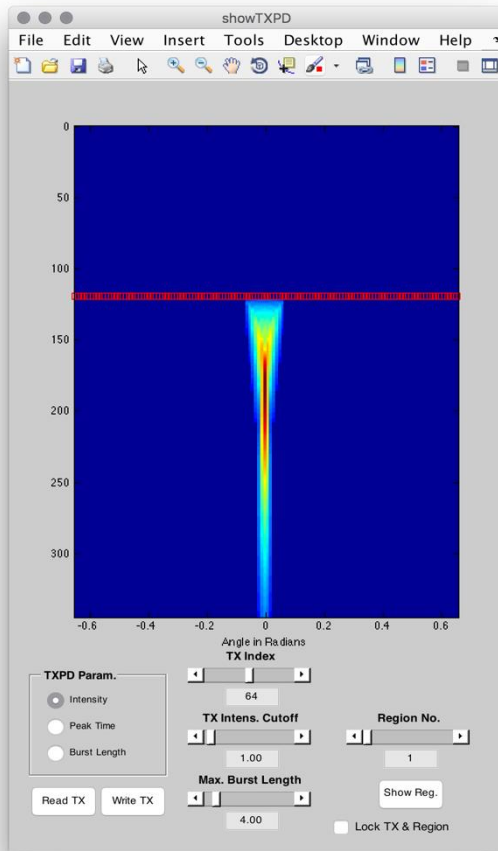


If the number of TX is equal to the number of Regions, a checkbox named “Lock TX & Region” will appear below the “Show Reg.” button. With “Lock TX and Region” checked, the TX or Region slider value will be consistent, so the user can inspect whether the transmit beam overlaps the region well. It’s important to ensure that the transmit beam overlaps the pixel region which is defined in the Recon. If the energy deposition on a certain pixel is not enough, the user may not see a clear image after reconstruction.

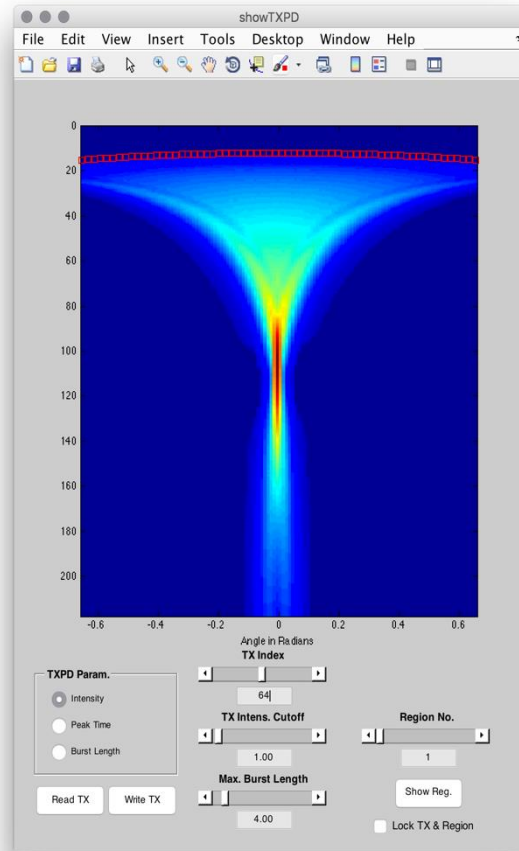
Visualization of polar coordinate

Because the PData definition supports Polar coordinate, showTXPD is able to visualize the transmit beam in Polar coordinate. As shown below, the unit of the x-axis is Angle in Radians instead of wavelength while Polar coordinate is used in PData.

C5-2v 128RvLnsPolar



P4-2v 128RvLnsPolar

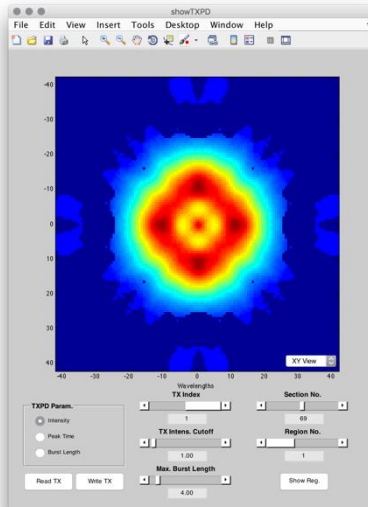


Three orthogonal view of 3-dimensional PData.Region

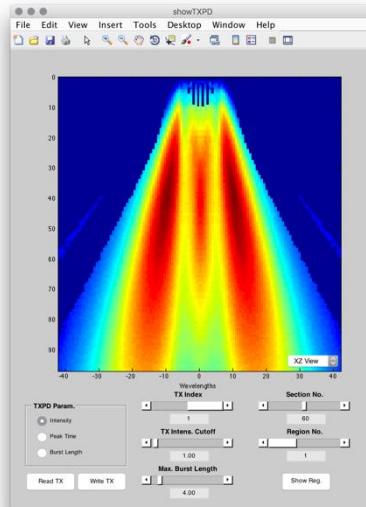
Typically, a 3D volume region is defined in PData when a 2D array is used. Currently, Verasonics provides two examples, list in Misc/Customized2Darry, about the definition of a 3D volume region. The showTXPD tool is able to display the region in three orthogonal views as well. As shown below, the default view (XY plane) is displayed after invoking showTXPD, along with a dropdown menu on the bottom right corner for the user to select different orthogonal view. The slider of Section No allows user to move the image along the third

dimension. By adjusting the TX intens. Cutoff, Max. Burst Length, and Region No., the user is able to get better ideas about how the transmit intensity overlaps with the pixel region.

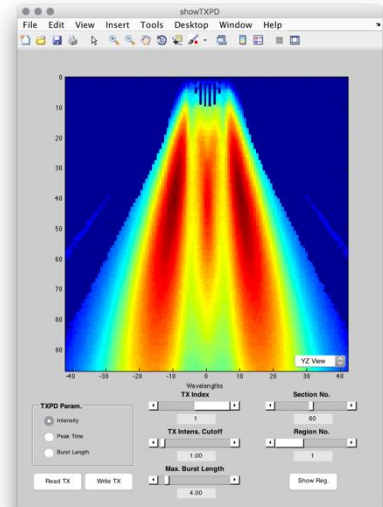
XY View



XZ View



YZ View



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